

PPOM1- OMM Lab #15: Diagnosis of the Pelvis Pre-Lab Lecture

Michael J. Terzella, D.O.

Session Objectives

- At the conclusion of the lab, the student should be able to:
- Identify and palpate clinically relevant landmarks of the pelvis/innominates.
- Comfortably perform all the steps related to diagnosis of the pelvis/innominates. Students will practice determining how to diagnose and name pelvic diagnoses.

Please note: The white slides are the ones that will be presented in the pre-recorded pre-lab lecture but ALL SLIDES are required for studying. Many overlap with the Basics of the Diagnosis of the Innominates and Pubic Bone lecture, which is a required lecture before viewing this slide presentation.

Osteopathic Evaluation of the Pelvis

- The diagnosis of pelvic dysfunction concentrates on the **ilial** and **pubic** components of the innominate
- The Standing Flexion Test indicates which side to name the innominate dysfunction (ie. ilio-sacral dysfunction).
- The ASIS compression test is also useful but is less specific, as it can indicate innominate *or* sacral dysfunction.
- Relative position of static landmarks allows one to correctly diagnose the pelvic somatic dysfunction
 - Static exam should be done with dominant eye over the area being examined

Landmarks and Tests for Pelvic Diagnosis

- Motion Testing:
 - Standing Flexion Test
 - ASIS Compression Test
- Static Landmarks:
 - Anterior superior iliac spine (ASIS)
 - Posterior inferior iliac spine (PSIS)
 - Pubic tubercles
 - Medial Malleoli
 - Ischial Tuberosities
- Diagnoses:
 - Innominate Rotations
 - Innominate Shears
 - Innominate Inflare/Outflare
 - Pubic Shears

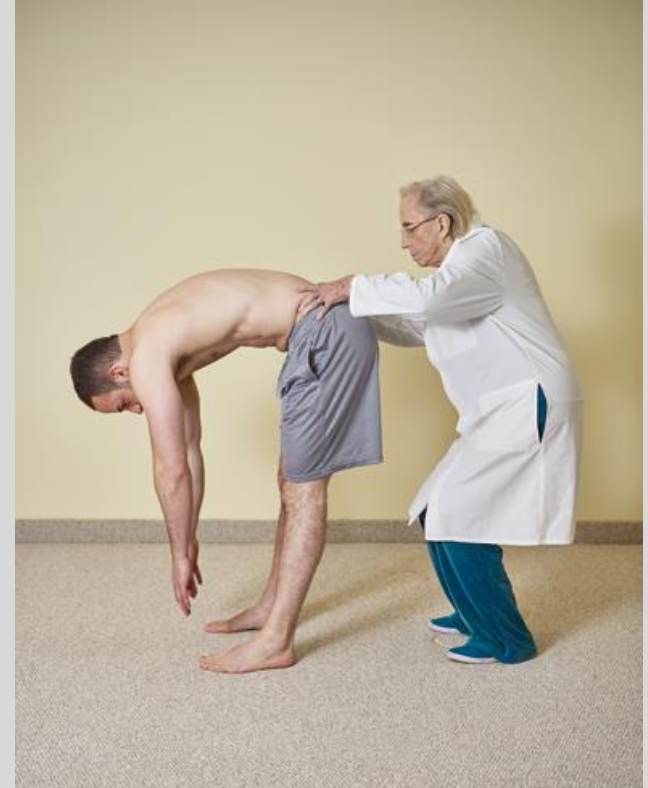
Sacral Landmarks

- Motion Testing:
 - Seated Flexion Test
 - ASIS Compression Test
- Static Landmarks:
 - Sacral Sulci
 - Inferior lateral angles of the sacrum (ILAs)
- Diagnoses:
 - We will go over how to make a diagnosis of the sacrum in another lab

Motion Testing

Standing Flexion Test (1 of 3)

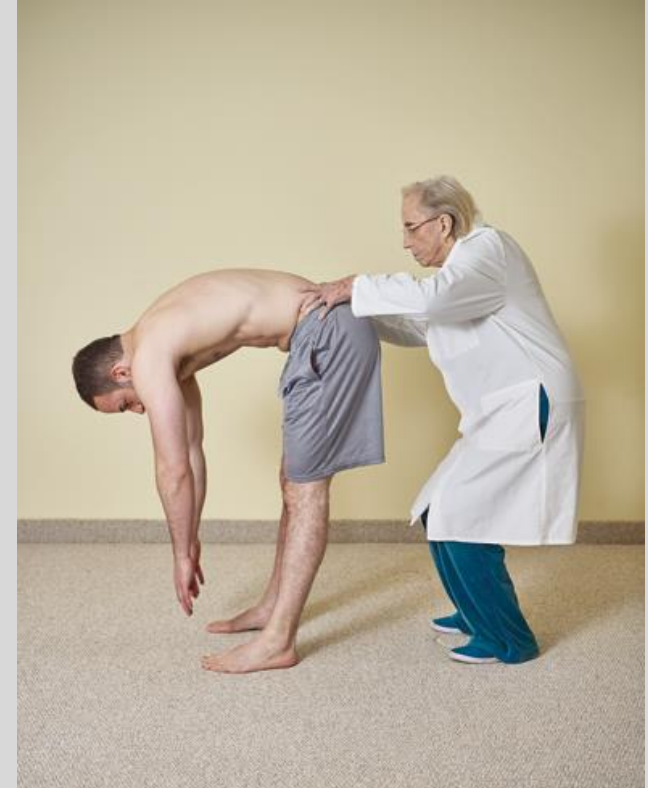
- Ask your patient to stand with his/her feet 6-8 inches apart
- Physician is eye level with the PSISs (kneeling or sitting)
- Place thumbs lightly at the inferior aspect of the PSISs (Must be on skin)
 - *Make sure your thumbs are horizontal (ie. parallel to the floor) for an optimal comparison



Standing Flexion Test (2 of 3)

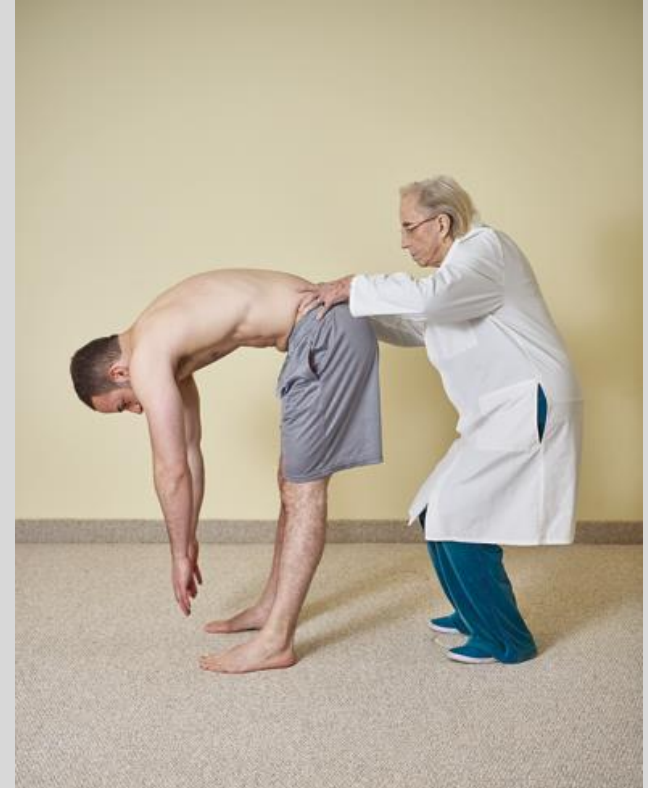
- Instruct Pt to slowly bend forward from the waist as if they were going to touch their toes without bending their knees
- As the Pt is flexing, the physician observes the movement of her thumbs on PSISs (test may be repeated to confirm)

****Keep thumbs firmly under the PSIS as the patient bends over. Do not allow them to roll over the PSIS**



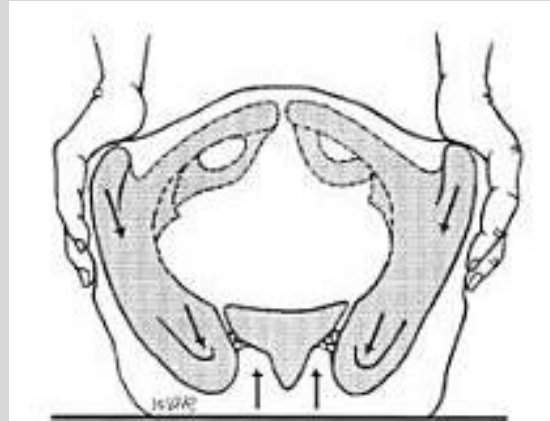
Standing Flexion Test (3 of 3)

- Positive Standing Flexion Test is indicated by the side where the thumb moves **“first and furthest”** (ie. more cephalad)
- **The side of the positive test is the side for which the dysfunction is named**
- **This test is specific for ilio-sacral dysfunction**



ASIS Compression Test (Pelvic Rock Test)

- For either ilio-sacral or sacro-iliac dysfunction
- Not specific for ilio-sacral dysfunction (ie. A positive test could mean either ilio-sacral or sacro-iliac dysfunction)



ASIS Compression Test (Pelvic Rock Test)

- Ask your patient to lie supine
- Place your thenar and hypothenar eminences along the patients ASISs bilaterally
- Introduce gentle but firm posterior and medial pressure toward the ipsilateral sacro-iliac joint on *one side at a time*
 - Slow
 - Allow the ilia to recoil anteriorly against gentle pressure
- Assess for *quality* of motion
 - One side usually offers more resistance



ASIS Compression Test (Pelvic Rock Test)

- Diagnosis: The side that offers more resistance with downward and medial pressure is the side of the ***positive*** test
- However, this test is nonspecific. It could be indicative of ilio-sacral dysfunction (i.e. pelvic) or of sacro-iliac dysfunction (i.e. sacral)
- Use to confirm findings

Static Landmarks

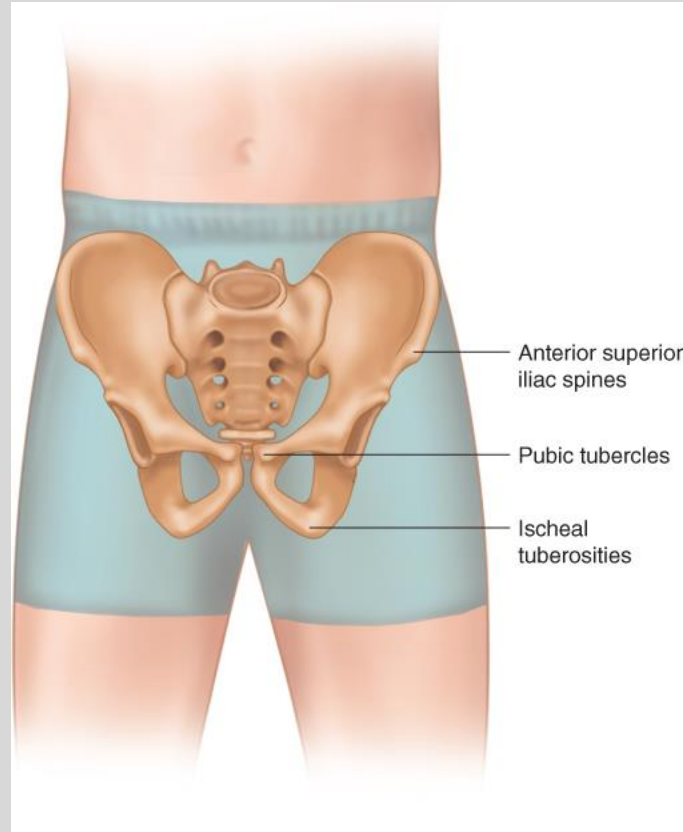
- Remember: the side of the positive **Standing Flexion Test** determines the side of the innominate dysfunction

Static Exam

- The first step is to *reseat the pelvis!*
 - This will make sure the pelvis is evenly placed on the table and that your patient is not leaning towards one side or the other
 - Ask your patient to bend his or her knees and place feet flat on table, then lift pelvis off table, then place their pelvis back down and straighten his or her legs onto the table
- Next, examine the Anterior and Posterior Landmarks

Pelvic Static Landmarks: Anterior

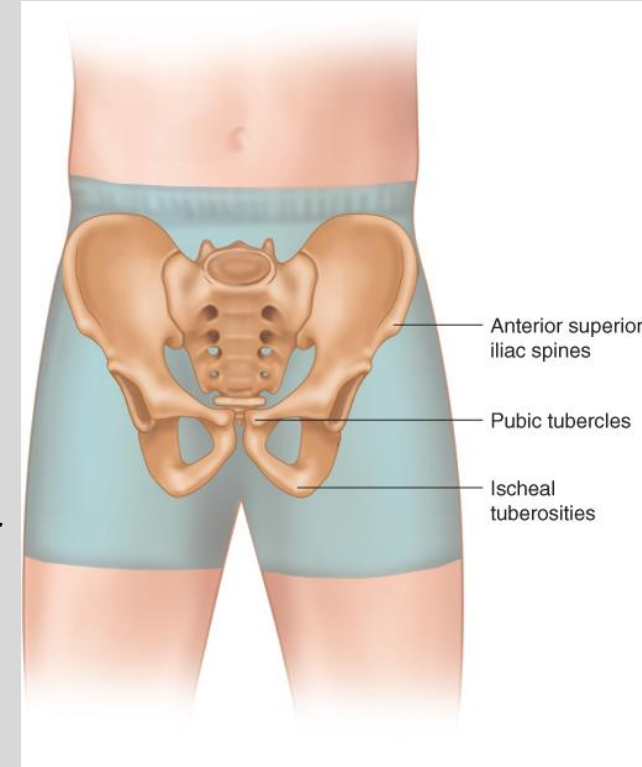
- Iliac Crests
- ASISs
- Pubic Tubercles
- Medial Malleoli



Pelvic Landmarks: Anterior

ASIS:

- Pt supine, physician standing with dominant eye closest to patient
- Stand so that your dominant eye can be over the vertical midline of the pt
 - Facing patient's head
 - May need to lean your side over the table to get your eye midline
- Lay the edge of your thumbs perpendicular to the inferior aspect of each ASIS – snugly – to determine which ASIS is relatively superior and which is inferior
- Then, using the umbilicus as a landmark, determine the relationship of the ASISs to the midline. (Which ASIS is closer, which is farther from midline?)



Pelvic Landmarks: Anterior

Pubic tubercles:

- Same beginning steps as above
- **Let your patient know that you intend to examine the lower part of their pelvis at the pubic bone (*consider showing the patient where this landmark is to be clear*).
- Starting at the level of the umbilicus, gently inch down with the heel of your hand (fingers pointing cephalad) until it meets the bony pubic symphysis
- Then gently place your thumb (or index finger) pads over the superior aspect of the pubic tubercles bilaterally to determine which side is relatively superior and which is inferior



Pelvic Landmarks: Anterior

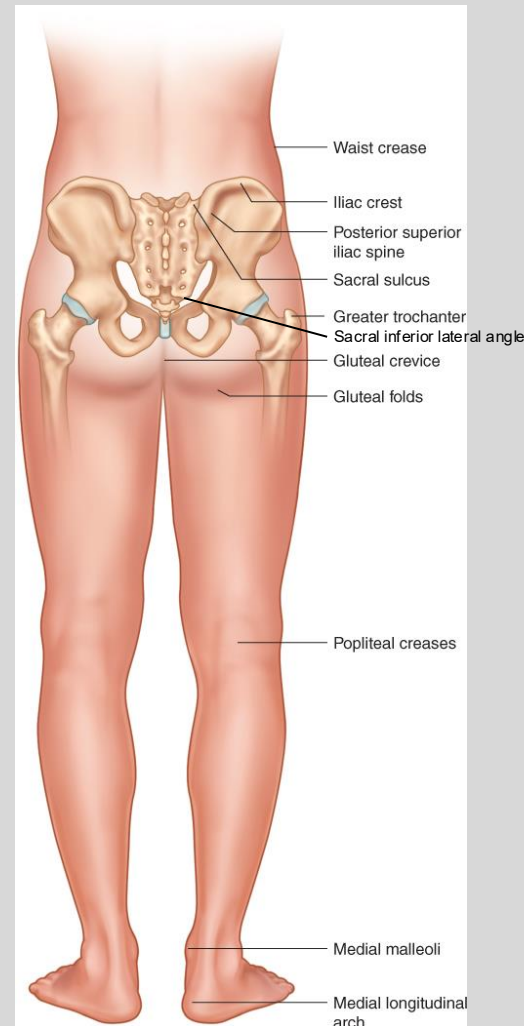
Medial Malleoli:

- If not done already, have the patient “reseat” their pelvis by bending knees, feet flat on the table, lifting pelvis in the air and putting back pelvis then legs back down
- Gently add traction inferiorly at both ankles to take out any slack
- Place your thumb pads just inferior to the medial malleoli bilaterally to determine which side is relatively superior and which is inferior



Pelvis: Static Landmarks: Posterior

- Iliac Crests
- PSISs
- Sacral sulci (dx sacrum)
- Inferior lateral angles of the sacrum (dx sacrum)
- Gluteal Folds
- Greater Trochanters
- Ischial Tuberosities
- Medial Malleoli



Pelvic Landmarks: Posterior

PSIS heights:

- Perform prone so you can see and feel after checking ASIS heights.
- Find the iliac crests.
- Place thumbs on the inferior slope of the PSISs bilaterally to determine which PSIS* is relatively superior and which is inferior.
- Alternatively (more challenging): One could determine the PSIS heights with your partner supine.
 - Reach around posteriorly with both hands to find the bilateral PSISs*
- *It is best to be on the skin to examine the PSISs



Practice

- Check Standing Flexion Test
- Check ASIS Compression Test
- Compare: ASISs, PSISs, pubic tubercles, and medial malleoli
- Write down findings (use next slide to mark your findings)
- Also, try to visualize the pelvis based on your findings and you will have an easier time to arrive at the correct somatic dysfunction diagnosis.

<u>Tests:</u>			<u>Circle Positive Side:</u>			
Standing Flexion Test			R	L	Neither	
ASIS Compression Test:			R	L	Neither	
<u>Landmarks:</u>	<u>Right Side</u> (circle one in each row)			<u>Left Side</u> (circle one in each row)		
ASIS	↑	↓	=	↑	↓	=
PSIS	↑	↓	=	↑	↓	=
Pubic Ramus	↑	↓	=	↑	↓	=
Medial Malleolus *If no leg length difference	↑	↓	=	↑	↓	=
ASIS Flare	In	Out	=	In	Out	=
Pubic Symphysis	Bulging or shallow cartilage					

Putting it all together- Naming pelvic SD

Tips:

- 1) Follow the steps in order
- 2) Be meticulous and consistent with your hand placement
- 3) Use dominant eye over the body's midline
- 4) **Name findings based on the side of the positive standing flexion test**
- 5) Visualize the dysfunctions. Use the hand models if needed to learn them well. (See the "Pelvic (Innominate) Diagnosis: Hand Model Supplementary Information" PowerPoint slide set. You may also refer to this video:
http://nyitmedicine.mediaspace.kaltura.com/media/PelvisA+Innominate+Diagnosis/1_9cz5q83s

Osteopathic Structural Diagnosis for the Pelvis

Standing Flexion Test

+

Supine Static Examination



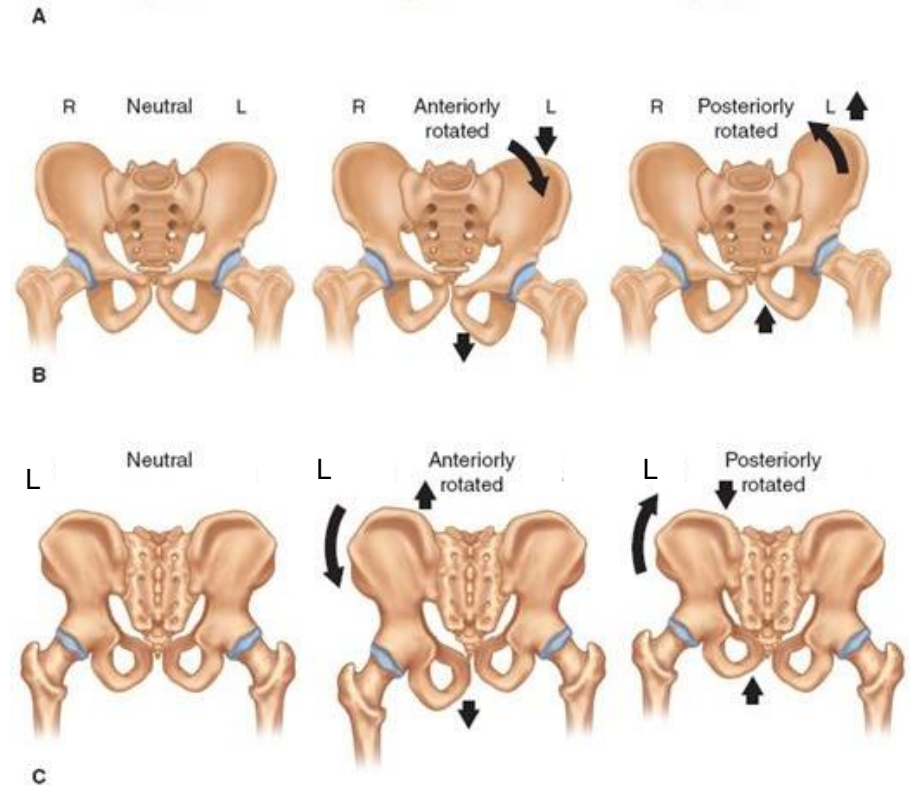
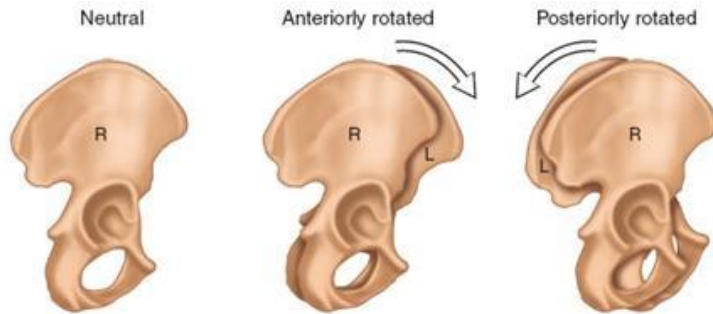
Somatic Dysfunction Diagnosis

Somatic Dysfunctions of the Pelvis

- Innominate Rotations
 - Right/left anterior rotation of the innominate
 - Right/left posterior rotation of the innominate
- Innominate Shears
 - Right/left superior shear of the innominate
 - Right/left inferior shear of the innominate
- Innominate Inflare/ Outflare
 - Right/left/bilateral outflare of the innominate
 - Right/left/bilateral inflare of the innominate
- Pubic Shears
 - Right/left superior pubic shear
 - Right/left inferior pubic shear
- Gapped Pubic Symphysis (abducted pubic symphysis)
- Compresses Pubic Symphysis (adducted pubic symphysis)

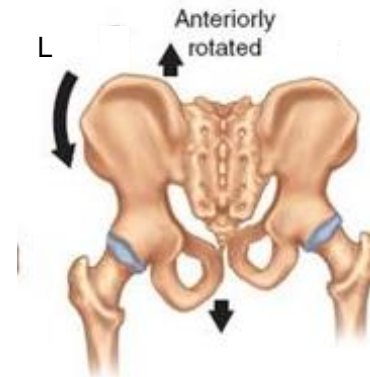
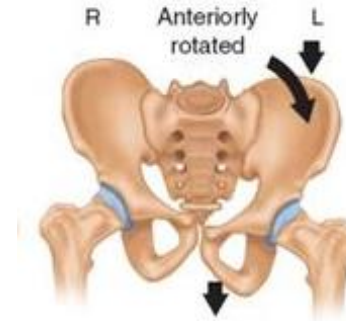
Innominate Rotations

- The entire innominate rotates either anteriorly or posteriorly in comparison to the sacrum



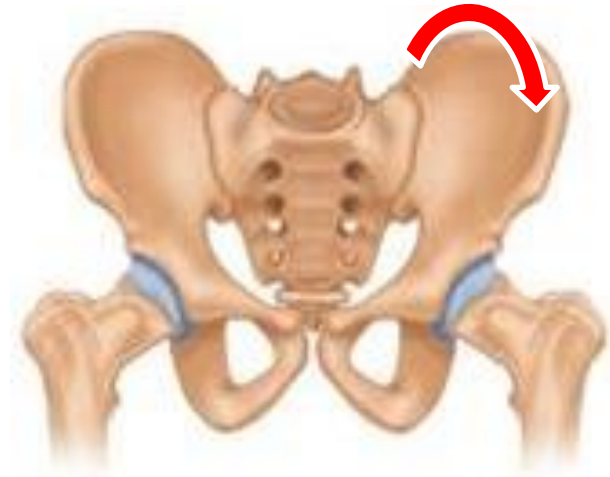
Left Anteriorly Rotated Innominate (findings will be opposite for a right anteriorly rotated innominate)

Standing Flexion Test	Positive on L
ASIS	L inferior
PSIS	L superior
Pubic Tubercle	L inferior or equal
Medial Malleolus	L inferior (if no leg length discrepancy)



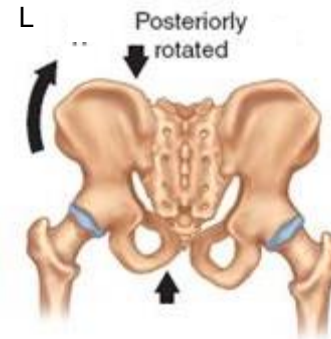
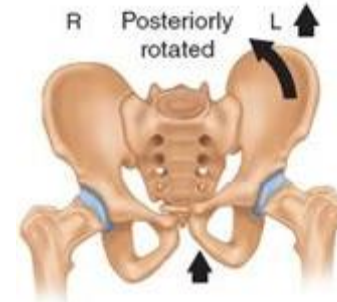
Anteriorly Rotated Innominate

- Subjective complaints may include:
 - Hamstring tightness
 - Sciatica secondary to piriformis dysfunction
- Palpatory findings:
 - Iliolumbar ligament tenderness
 - Freedom of anterior rotation motion testing
 - Restriction to posterior rotation motion testing



Left Posteriorly Rotated Innominate (findings will be opposite for a right posteriorly rotated innominate)

Standing Flexion Test	Positive on L
ASIS	L superior
PSIS	L inferior
Pubic Tubercle	L superior or equal
Medial Malleolus	L superior (if no leg length discrepancy)



Posteriorly Rotated Innominate

Subjective complaints may include:

- Inguinal/groin pain secondary to rectus femoris dysfunction (*WHY?*)
- Medial knee pain secondary to sartorius dysfunction (*WHY?*)

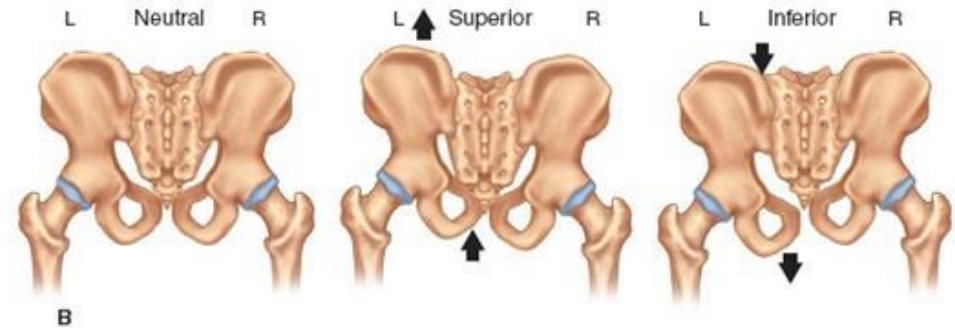
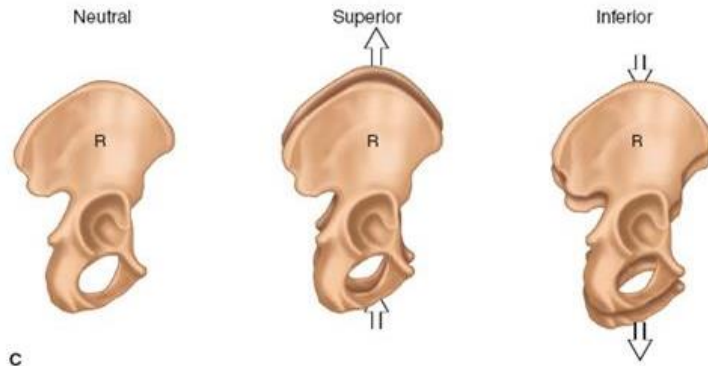
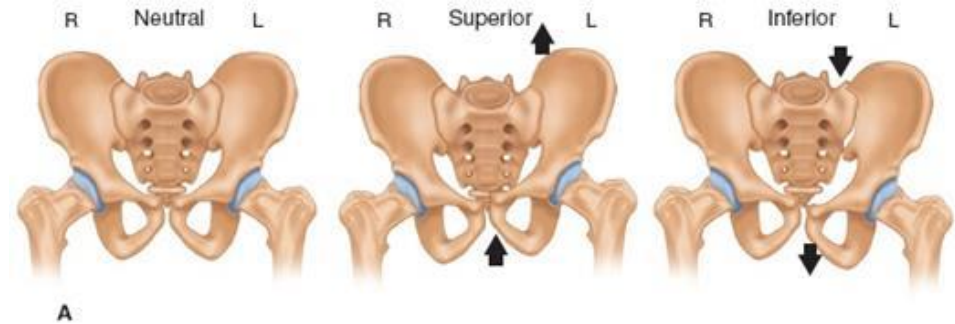
Palpatory findings:

- Inguinal tenderness
- Freedom of posterior rotation motion testing
- Restriction to anterior rotation motion testing



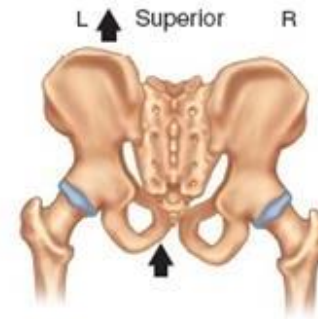
Innominate Shears

- The entire ilium is shifted inferiorly or superiorly in relation to the sacrum
- Usually occur as a result of **trauma**



Left Superior Iliac Shear

Standing Flexion Test	Positive on L
ASIS	L superior
PSIS	L superior
Pubic Tubercle	L superior (typically)
Medial Malleolus	L superior (if no leg length discrepancy)



Superior Innominate Shear



Subjective complaints may include:

- Pelvic pain

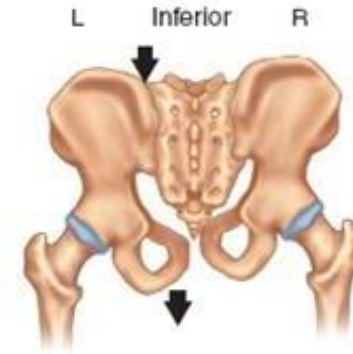
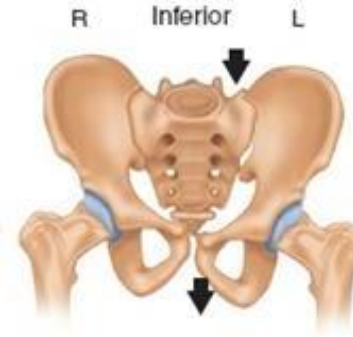
Palpatory findings:

- May have TTC (tissue texture changes) at ipsilateral sacroiliac (SI) joint or pubes
- Motion testing has freedom of superior translation

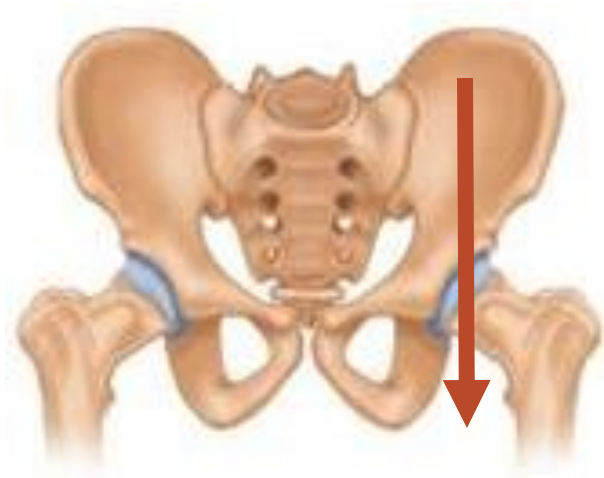
What might cause a patient to have this dysfunction?

Left Inferior Innominate Shear

Standing Flexion Test	Positive on L
ASIS	L inferior
PSIS	L inferior
Pubic Tubercle	L inferior (typically)
Medial Malleolus	L inferior (if no leg length discrepancy)



Inferior Innominate Shear



Subjective complaints may include:

- Pelvic pain

Palpatory findings:

- May have TTC (tissue texture changes) at ipsilateral SI joint or pubes
- Motion testing has freedom of inferior translation

What might cause a patient to have this dysfunction?

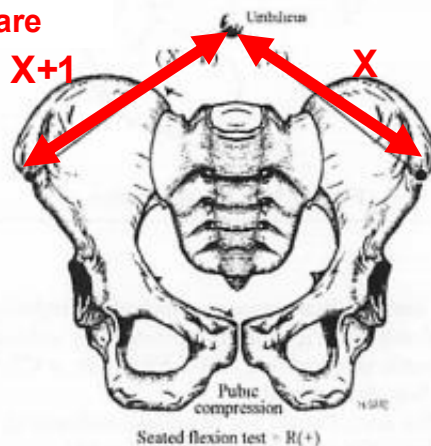
Innominate Outflare/Inflare

OUTFLARES

- ASIS – one (or both) apparently further from the midline
- Associated with posterior rotation due to shape of SI joint

- *Subjective complaints: may have pelvic or SI joint pain*

Right outflare



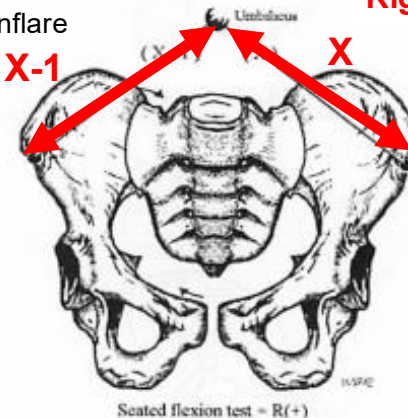
INFLARES

- ASIS – one (or both) apparently closer to the midline
- Associated with anterior rotation due to shape of SI joint

Right inflare

Right inflare

X-1



Pubic Shears

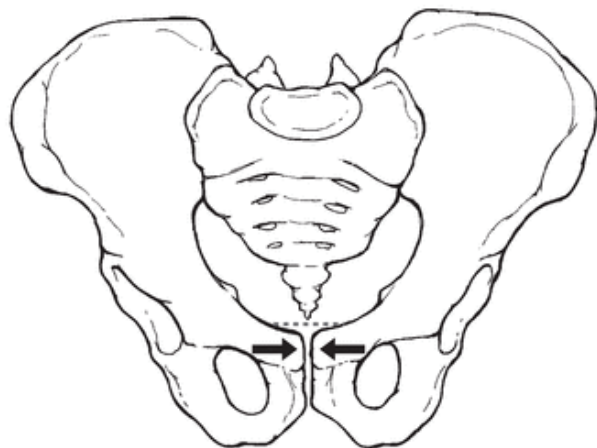
- The ASIS, PSIS heights appear **equal**
- Can have superior or inferior pubic shear
- **Named for the side of the positive standing flexion test**
- If the pubic tubercle is higher on the side of the positive standing flexion test, it is a superior pubic shear on that side
- If the pubic tubercle is lower on the side of the positive standing flexion test, it is an inferior pubic shear on that side
- Associated with pubic pain related to trauma, childbirth, possibly cystitis, enuresis, prostatitis, etc



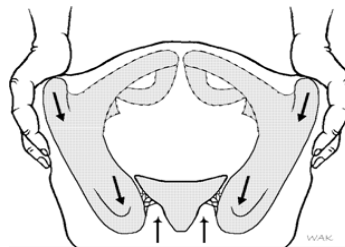
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Pubic Compression Dysfunction (AKA **AD**ducted pubic symphysis) -

Also called pubic **add**uction. A somatic dysfunction in which the pubic bones are forced toward each other at the pubic symphysis. This dysfunction is characterized by tenderness to palpation over the pubic symphysis, lack of apparent asymmetry, but associated with restricted motion of the pelvic ring (from FOM4 Glossary)



Atlas of Osteopathic Tech 3e, Ch 10



Foundations 3e, Ch. 41



Standing flexion test = Bilaterally(+) (False negative)

****ASIS compression test may be falsely neg or may lateralize to one side; please keep in mind this test may indicate innominate OR sacral Dysfunction, but is useful as a screening test.**

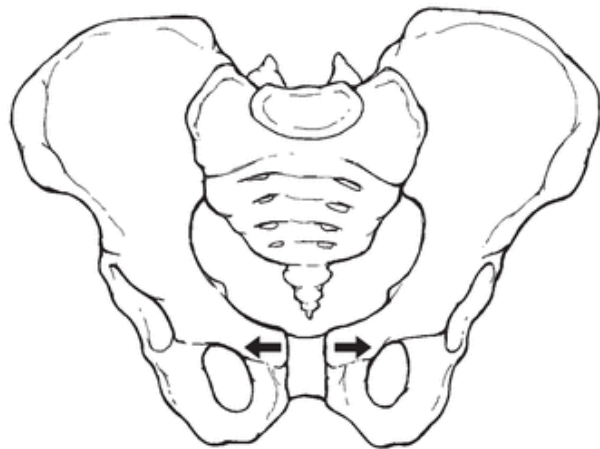
****Standing flexion test may be falsely neg or may lateralize to one side. It is considered as indicative of innominate (which pubes are part of) dysfunction.**

Pubis Compression Dysfunction

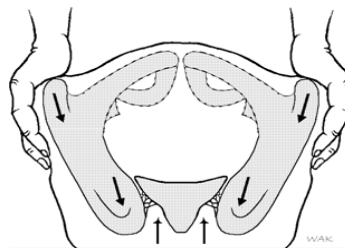
- Pubic symphysis compression (also called pubic adduction) occurs when the pubic bones are forced toward each other at the symphysis.
- Suspicion of dysfunction by history (trauma, pregnancy, delivery)
- Standing flexion test may be positive on one or both sides (false negative)
- There is no apparent asymmetry of the pubic bones(The ASIS, PSIS heights appear **equal**).
- There is bilateral tenderness to palpation at the symphysis and often on each pubic rami as well.
- There is an associated restriction of motion of the pelvic ring (ASIS compression restricted B/L).
- Palpable bulging of the symphyseal cartilage
- May have urinary tract symptoms

Pubic Gapping Dysfunction (AKA **AB**ducted pubic symphysis) -

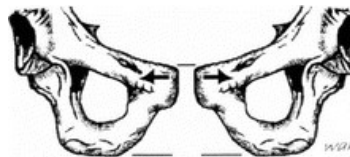
Also called pubic **ab**duction. A somatic dysfunction in which the pubic bones are pulled away from each other at the pubic symphysis. This dysfunction is frequently seen in women following childbirth (from FOM4 Glossary).



Atlas of Osteopathic Tech 3e, Ch 10



Foundations 3e, Ch. 41



Standing flexion test = Bilaterally(+) (False negative)

****ASIS compression test may be falsely neg or may lateralize to one side; please keep in mind this test may indicate innominate OR sacral Dysfunction, but is useful as a screening test.**

****Standing flexion test may be falsely neg or may lateralize to one side. It is considered as indicative of innominate (which pubes are part of) dysfunction.**

Pubis Gapping Dysfunctions

- Pubic symphysis gapping (also called pubic abduction) occurs when the pubic bones are forced away from each other at the symphysis.
- Suspicion of dysfunction by history (trauma, pregnancy, delivery)
- Standing flexion test may be positive on one or both sides (false negative)
- There is no apparent asymmetry of the pubic bones (The ASIS, PSIS heights appear **equal**).
- Sulcus deeper than normal at the pubic symphysis
- Tender pubic symphysis and B/L on the tubercles and often on the rami
- May have urinary tract symptoms
- Gapping or subluxation of the pubic symphysis can occur after childbirth, after pelvic fractures, or other traumas. Pelvic x-rays clearly document the degree of gapping. These are usually quite painful, and not easily treated.

Documentation- It's important!

It is important to be able to efficiently examine for, name, and record somatic dysfunction. These are the proper ways to write the pelvic dysfunctions – examples are for right-sided dysfunctions:

Right Anteriorly rotated innominate

Right posteriorly rotated innominate

Right superior innominate shear

Right inferior innominate shear

Right innominate inflare

Right innominate outflare

Right superior pubic shear

Right inferior pubic shear

Compressed pubis or adducted pubis

Gapped pubis or abducted pubis

Case Presentation:

- Please read over the case and try to answer the questions.
- The Case answers are posted separately with the rest of the lab material.

Case Presentation:

A 20-year-old college equestrian team member with a history of depression presents to your office complaining right lower back and hip pain since she fell from her horse during practice one month ago. She says her right foot got caught in the stirrup when she fell off. She was evaluated in the emergency department the day of the injury. Fortunately, she did not break any bones or sustain any other injuries. The pain is located in the right buttock, pubic and inguinal area as well as in her lumbar spine bilaterally. The pain is 8/10 in severity and it is worse when she tries to ride her horse. Consequently, she has not been able to ride for the past month and has been feeling very “down” lately. She denies any numbness, tingling or weakness in her legs.

- **Allergies:** None
- **Medications:** Fluoxetine 20 mg every morning
- **Past Medical History:** Depression
- **Past Surgical History:** none
- **Family history:** Both parents are alive and well
- **Social History:** non-smoker. Does not use any drug for recreational pleasure. Does not drink alcohol.

Physical Exam:

Heart: normal S1, S2. No murmurs, rubs or gallops

Lungs: Clear to auscultation B/L

Abdomen: Soft, non-tender, non-distended, no guarding, rigidity or rebound tenderness

Neurologic exam: lower extremity motor 5/5 bilaterally; L4, S1 deep tendon reflexes 2/4 bilaterally

Structural exam: B/L paravertebral muscle spasm (moderate), bilateral psoas spasm (severe), + standing flexion test on the right, right ASIS higher than left, right PSIS higher than the left, right pubic tubercle higher than the left (pelvic dysfunction severe). Bilateral hamstring hypertonicity R>L (moderate).

What is the pelvic somatic dysfunction?

+ standing flexion test on the right, right ASIS higher than left, right PSIS higher than the left, right pubic tubercle higher than the left (pelvic dysfunction severe).

What are the patient's somatic dysfunctions?

Right Superior innominate Shear

Psoas spasm

Paravertebral muscle spasm

Hamstring hypertonicity

Summary

- Osteopathic evaluation of the Pelvis includes putting together the Motion Testing and Static Landmarks exam to name somatic dysfunctions of the pelvis
- Remember to name landmarks on the side of the positive standing flexion test
- Pubic dysfunctions are named only if the ASIS and PSIS are even
- Pelvic somatic dysfunction can cause musculoskeletal as well as organ related complaints, so they are useful to check for and treat, if dysfunction is found

References

An Osteopathic Approach to Diagnosis and Treatment, 4e
Eileen L. DiGiovanna, Christopher J. Amen, Denise K. Burns:

Chapter 57: Pelvic and Sacral Anatomic Considerations

<https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=248813935&bookid=2969>

Ch 59 Evaluation of the Pelvis

<https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=248814094&bookid=2969>

Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research, 5e

Ch 37 and Glossary of Osteopathic Terminology

<https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/book.aspx?bookid=3456>

Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research, 4e

Ch 41b

<https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=209552184&bookid=2582>

Atlas of Osteopathic Techniques, 4e

Ch10

<https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=252634405&bookid=3202#252634739>

Atlas of Osteopathic Techniques, 3e

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<https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=123844140&bookid=1629#144162655>

Lab Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>